Exp.No: 02 DETERMINATION OF CURRENT IN CIRCUIT USING MESH ANALYSIS

Date: 21/2/26

Pg No.

AIM:

To determine the current in circuit using mesh analysis both theoretically and practically for a given DC circuit.

APPARTUS REQUIRED:

SL.NO	APPARATUS	SPECIFICATION	QUANTITY
1	RPS	(0-30)V	1
2	Ammeter	(0-10mA) MC	3
3	Resistors	330Ω, 470Ω, 560Ω	Each 1
4	Bread Board	-	1

PROCEDURE:

1) Give connections as per the circuit diagram.
2) Switch ON the supply, vary the RPS and set a particular input voltage.
3) Note down the readings of ammeters and voltmeters and tabulate them.
4) Vary the RPS for different input voltages and note down the readings of all the meters.
5) Reduce the RPS to its minimum value and switch OFF the supply.
6) Using the tabulated values, verify Kirchoff's laws. Practically, and verify it theoretically.

TABULAR COLUMN

Parameter	Theoretical	Practical
I_1	1.22	1.3m A
I_2	-4.29	-4.3 mA
$-I_2$	5.51	5.5mA

CALCULATIONS :

$$\begin{bmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

$R_{11} = 330 + 470\,\Omega = 800\,\Omega$ $\quad R_{12} = 470\,\Omega$

$R_{21} = 470\,\Omega$ $\quad R_{22} = 1030\,\Omega$

$$\begin{bmatrix} 800 & -470 \\ -470 & 1030 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 3 \\ -5 \end{bmatrix}$$

$\Delta = 800 \times 1030 - (470)^2 = 603100$

$$\Delta_1 = \begin{bmatrix} 3 & 470 \\ -5 & 1030 \end{bmatrix} = 3090 - 2350 = 740$$

$\Delta_2 = -4000 + 1410 = -2590$

$$I_1 = \frac{\Delta_1}{\Delta} = \frac{74\cancel{0}}{60310\cancel{0}} = 1.22\,mA$$

$$I_2 = \frac{-259\cancel{0}}{60310\cancel{0}} = -4.29\,mA$$

$$I_1 - I_2 = 1.22 - (-4.29) = 5.51\,mA$$

MARK SPLITUP:

S.No	Description	Marks Alloted	Marks Obtained
1	Experimental setup/ Connection	20	20
2	Observation / Reading	20	20
3	Conclusion / result	20	20
4	Viva	20	20
5	Record	20	20
		Total	100

100

10/2/26

Result:

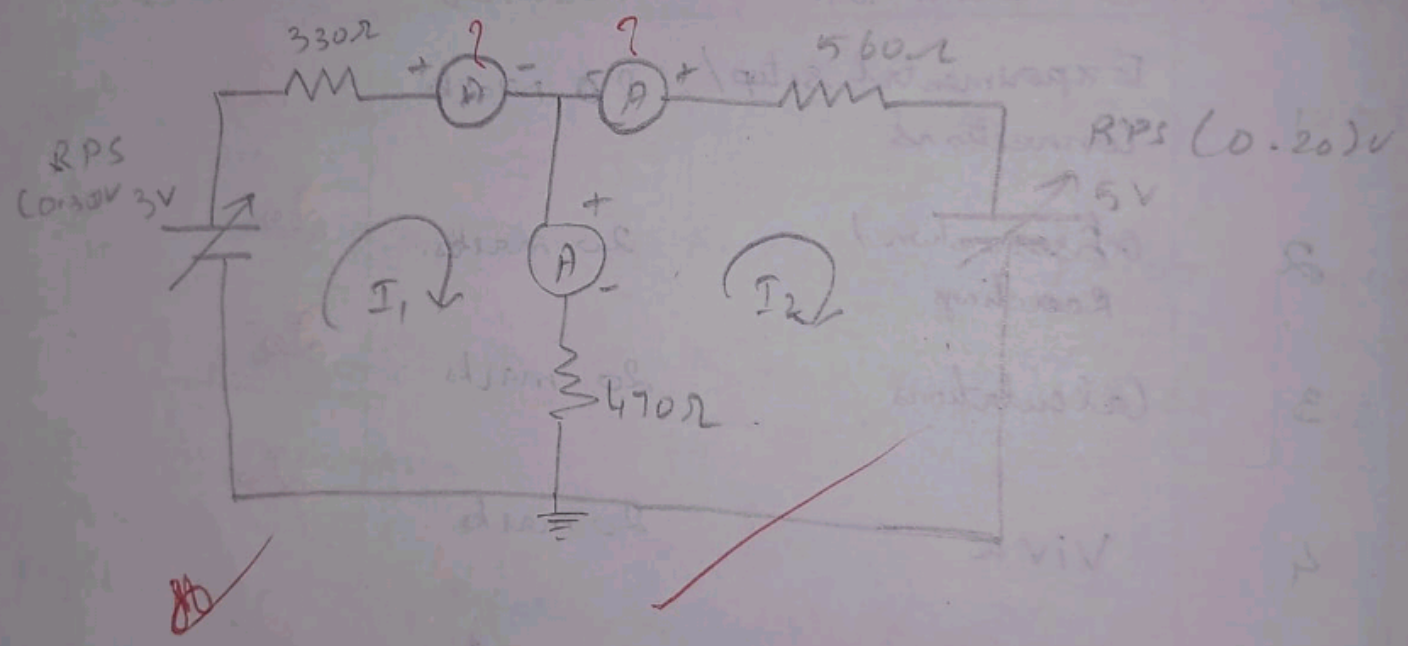
Thus, the Mesh analysis is verified practically and theoretically.

The resultant currents for 3V | 5 V supply are:

a) The current across 33Ω resistor is 1.3m A.

b) the current across 560Ω resistor is -4.3m A.

c) The current across 470Ω resistor is 5.5m A.

CIRCUIT DIAGRAM.

MESHI CIRCUIT